



GEN AI FRAUD ANALYST

ARCHITECTURE FLOW

STEP-BY-STEP SYSTEM FLOW



EXPLANATION OF EACH STEP

■ Dataset Input

- The project begins with a **CSV dataset** containing raw data.
- Example data may include:
- categories
- numerical values
- attributes used for analysis
- This dataset is the **input of the system**.

■ Data Loading

- The dataset is loaded using **Python**.
- Libraries used:
- **Pandas** → for data handling
- **NumPy** → for numerical operations
- Purpose:
- Read dataset
- Convert into DataFrame format

EXPLANATION OF EACH STEP

■ **Data Cleaning**

- Before analysis, the dataset is cleaned.
- Tasks performed:
 - Removing missing values
 - Handling duplicates
 - Formatting columns
 - Fixing incorrect data
- This step ensures **accurate analysis**.

■ **Exploratory Data Analysis (EDA)**

- EDA is performed to understand the dataset.
- Operations include:
 - statistical summaries
 - correlation checks
 - feature relationships
 - distribution analysis
- EDA helps identify **patterns and trends in data**.

EXPLANATION OF EACH STEP

- **Data Visualization**

- Graphs are generated using:

- **Seaborn**

- **Matplotlib**

- Examples of charts used:

- Bar Plot

- Scatter Plot

- Histogram

- Heatmap

- Line Plot

- These visualizations help interpret complex datasets easily.

- **Output / Insights**

- The final output of the project includes:

- visual graphs

- pattern analysis

- insights from the dataset

- These insights help users **understand trends and relationships in the data.**